PROJECT PROPOSAL

Vashasetu

Turn any regional dialect into Standard Bangla — by voice, instantly.

Software Engineering
CSE 355

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Problem Statement

The Regional Dialect Problem: Bangladesh's regional dialects — Chattogram, Sylheti, Noakhali, Barisal, Rangpur, and others — diverge sharply from Standard Bangla in vocabulary and pronunciation. Existing digital services, from government helplines to translation apps, are built around Standard Bangla, effectively excluding dialect-speaking populations, particularly rural and elderly users, from services that assume a fluency they do not have.

The Vashasetu Software Solution: Vashasetu is a voice-first, full-stack translation platform designed to solve this problem by combining speech recognition, a fuzzy-matching dictionary engine, and speech synthesis into a single, seamless pipeline. A user speaks in their own dialect; the system converts speech to text, matches it against a growing dictionary using approximate-pattern matching (to handle non-standardized spelling), and returns Standard Bangla — both as text and as spoken audio.

Sectoral & Social Impact: By giving dialect speakers a voice-based alternative to typing and reading Standard Bangla, Vashasetu reduces digital exclusion, supports rural and elderly users accessing public services, and creates a reusable, community-extensible dictionary of regional vocabulary.

Academic Scope & Execution Horizon: The project is scoped for execution across a structured academic timeline by a two-member development team, culminating in a functional web demonstration, a documented dictionary and matching engine, and a final academic presentation.

1. Project Objectives

1.1 General Objective

The primary objective of the Vashasetu project is to design, develop, test, and deploy a voice-enabled, full-stack translation web application within a structured academic framework, providing accurate, two-way translation between regional Bangla dialects and Standard Bangla.

1.2 Specific Measurable Objectives

1. Two-Way Translation Engine: Build a dictionary and fuzzy-matching system that translates regional dialect text to Standard Bangla and back, tolerating spelling variation.

2. Voice Input Pipeline: Integrate speech-to-text (Web Speech API / Whisper) so users can speak in their dialect instead of typing.

3. Voice Output Pipeline: Integrate text-to-speech so translated Standard Bangla is read aloud, keeping the interaction conversational.

4. Community Contribution Workflow: Allow users to submit new dialect words, with admin review before they are added to the live dictionary.

5. Reliable Hybrid Input: Guarantee a typed-text fallback on every screen where voice input is offered, so the system remains usable if microphone access fails.

2. Project Scope & System Boundary

2.1 In-Scope (Minimum Viable Product - MVP)

- Dialect ↔ Standard Bangla Translation Subsystem: Text and voice translation, word-by-word breakdown.
- Voice Interface: Microphone-based speech input and spoken audio output (STT/TTS).
- Dictionary & Fuzzy-Matching Engine: The backend engine that resolves spelling variation across dialects.

- Contributor Panel: Community word submission with administrator review before publication.
- Translation History: Per-user log of past translations.

2.2 Out-of-Scope (Explicit Exclusions)

- Full offline speech recognition (requires internet connectivity).
- Support for languages other than Bangla and its regional dialects.
- Real-time two-person conversation mode.
- Native mobile applications (initial release is web-first).

2.3 Target User Roles & Responsibilities Matrix

- Dialect Speaker (Applicant-equivalent role): Speaks or types dialect input, views translated output, contributes new words.
- Administrator: Reviews and approves community word contributions, manages the dictionary, resolves reported mistranslations.

3. Comprehensive Feasibility Study

3.1 Technical Feasibility

Technical feasibility is high. The proposed stack — PHP, MySQL, the Web Speech API, and OpenAI Whisper — represents mature, well-documented technologies already within the team's coursework. MySQL is sufficient for the dictionary and fuzzy-matching lookup tables at the scale required for this MVP.

3.2 Economic Feasibility

Academic / Demonstration Cost: Free-tier speech APIs and low-cost shared hosting keep the total academic demonstration cost minimal.

Potential Production Operational Cost: Market research (Section 6) indicates an estimated annual cost of approximately 7,700 BDT for domain, hosting, and speech-API usage at low-to-moderate traffic.

3.3 Operational Feasibility

Vashasetu replaces a completely manual process — dialect speakers relying on human translators or guesswork — with a self-service tool, requiring no special training for end users.

3.4 Schedule Feasibility

The project schedule spans a structured academic period divided into six sequential phases (Section 4), with documentation running in parallel throughout to reduce end-of-project workload.

4. Work Breakdown Structure (WBS)

4.1 Hierarchical WBS Task Breakdown

1.0 Requirement Gathering

- 1.1 Interview regional dialect speakers and collect sample phrases.
- 1.2 Finalize functional requirements.

2.0 Design

- 2.1 Produce UI/UX wireframes in Figma.
- 2.2 Design database schema and system architecture.

3.0 Dictionary Compilation

- 3.1 Collect dialect-to-standard word pairs across regions.
- 3.2 Structure fuzzy-matching rules and the admin review workflow.

4.0 Voice Integration

- 4.1 Integrate the STT engine (Web Speech API / Whisper).
- 4.2 Integrate the TTS engine and test recognition accuracy across dialects.

5.0 Development

- 5.1 Build the Translate, History, and Contribute screens.
- 5.2 Build the backend API and database integration.

6.0 Testing & Deployment

- 6.1 Execute unit and user-acceptance testing.
- 6.2 Deploy to Hostinger and finalize documentation.

4.2 WBS Responsibility Matrix

WBS ID	Work Package	Summary Deliverable	Responsible
1.0	Requirement Gathering	Finalized functional requirements	Both Members
2.0	Design	Wireframes, DB schema, architecture	Both Members
3.0	Dictionary Compilation	Seed dictionary, fuzzy-match rules	Muskan (Backend)
4.0	Voice Integration	Working STT/TTS pipeline	Muskan (Backend)
5.0	Development	Functional screens + API integration	Liza (Frontend)
6.0	Testing & Deployment	Tested, deployed application	Both Members

5. Gantt Chart & Project Timeline

5.1 Project Schedule & Dependency Matrix

Task Description	Start	End	Duration	Dependencies
1. Requirement Gathering	Week 1	Week 1	**1 Week**	None
2. Design	Week 2	Week 2	**1 Week**	Task 1
3. Dictionary Compilation	Week 2	Week 3	**2 Weeks**	Task 1
4. Voice Integration	Week 3	Week 4	**2 Weeks**	Task 3
5. Development	Week 4	Week 6	**3 Weeks**	Task 2, 4
6. Testing & Deployment	Week 7	Week 8	**2 Weeks**	Task 5

6. Budget & Financial Plan

6.1 Market Research Methodology & Sources

Market pricing was retrieved directly from provider pricing pages in August 2026, including Hostinger, Namecheap, Google Cloud, and OpenAI. Calculations assume a 1 USD ≈ 120 BDT exchange rate where relevant.

6.2 Itemized Market Research Resource Budget Table

Item	Specification / Tier	Cost (BDT)	Billing / Source	Requirement
Web Hosting	Hostinger Shared Plan	**2,500 BDT**	Annual / Hostinger	Production
Domain (.com)	1-Year Registration	**1,200 BDT**	Annual / Hostinger	Production
Speech-to-Text API	Google Cloud (buffer beyond free tier)	**1,500 BDT**	Usage-based / Google Cloud	Academic & Prod
Text-to-Speech API	Google Cloud (buffer beyond free tier)	**1,000 BDT**	Usage-based / Google Cloud	Academic & Prod
UI/UX Design Assets	Icon & illustration packs	**800 BDT**	One-time	Academic & Prod
Contingency (~10%)	Buffer for unforeseen costs	**700 BDT**	One-time	Academic & Prod
Total Estimated Budget		**7,700 BDT**	*≈ $64 USD*	

6.3 Financial Summary

The total estimated first-year budget is 7,700 BDT (approx. $64 USD), as itemized above. Figures will be re-verified against live provider pricing before final submission.

7. System Specifications & Technical Architecture

7.1 Database Schema & Relational Structure

The backend is anchored on a normalized MySQL schema. Primary tables include users (authentication and preferred dialect), dialect_words (dialect word, standard meaning, dialect type, review status), translation_history (per-user translation log), and contributions (community-submitted words pending admin review).

7.2 Core API Endpoint Specifications

- POST /api/translate: Accepts dialect text or transcribed speech and returns the matched Standard Bangla translation.
- POST /api/stt: Forwards audio input to the speech-to-text engine and returns transcribed dialect text.
- POST /api/tts: Converts translated Standard Bangla text into spoken audio output.
- POST /api/contribute: Submits a new dialect word for admin review before it is added to the dictionary.

7.3 Security & Data Protection Controls

Passwords are stored using bcrypt hashing. All database queries are parameterized to prevent SQL injection. Community-contributed words are held in a pending state and are never published to the live dictionary without administrator approval.

8. References

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